Monocular Visual Odometry To Visual-Inertial Odometry on TUM VI

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Abstract

This work implements a classical geometry-based Monocular Visual Odometry (VO) pipeline and extends it to Visual-Inertial Odometry (VIO) using IMU preintegration and sliding-window optimization. VO suffers from **scale ambiguity**. VIO fuses inertial data to recover metric scale. Evaluated on TUM VI (Room2, Corridor3, Outdoors5), VO achieves ATE of 0.72 m, 1.13 m, 1.08 m after Sim(3) alignment, with low RPE (0.015 m/100m). VIO with SE(3) alignment shows scale is not fully recovered due to a simplified gradient-based optimizer.

Introduction & Problem Formulation

Visual Odometry (VO): estimates camera motion from images alone.
Monocular VO limitation: scale ambiguity. From epipolar geometry, the Essential matrix E gives $\mathbf{t}$ only up to an unknown scale:

$$\mathbf{x}'^{\top} E\mathbf{x} = 0, \quad E = [\mathbf{t}]_{\times} R, \quad \|\mathbf{t}\| = 1$$

A trajectory of 1m or 100m produces identical image projections. **VIO:** fuses IMU (accelerometer + gyroscope) to provide metric scale via $F = ma$.

Pipeline Overview

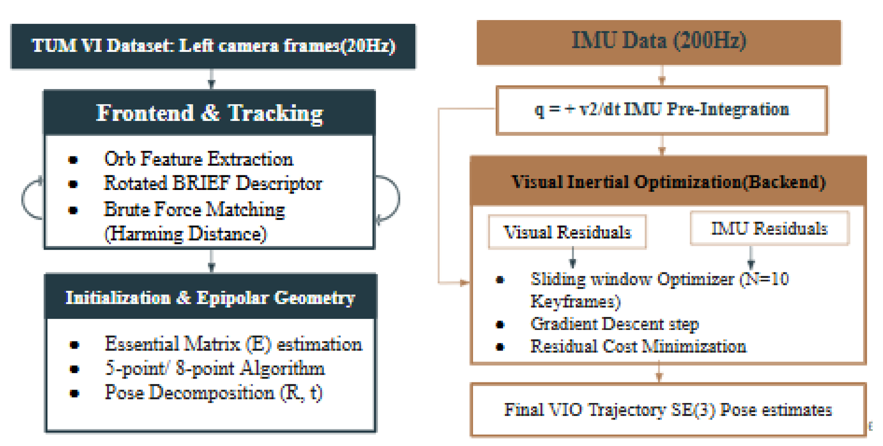


Fig. 1: Pipeline VO/VIO

Research Gap & Contributions

- Mature systems (ORB-SLAM3, VINS-Mono) are complex black boxes.
- This work **deliberately omits** BA, loop closure, bias estimation.
- **Contributions:** VO baseline, VIO with preintegration, ablation study.

Methodology — VO Frontend

ORB features (2000 keypoints). Brute-force Hamming + Lowe's ratio test + cross-check.

$$\frac{d(d_i, d_{j1})}{d(d_i, d_{j2})} < 0.75$$

Essential Matrix via RANSAC (p=0.999, thresh 1px):

$$\mathbf{x}'^{\top} E\mathbf{x} = 0, \quad E = [\mathbf{t}]_{\times} R$$

Trajectory: $T_k = T_{k-1} \cdot T_{rel}$, scale ambiguous.

Methodology — VIO & IMU Preintegration

IMU preintegration (Forster et al.):

$$\Delta\mathbf{R}_{ij} = \prod \mathrm{Exp}((\omega_k - \mathbf{b}_\omega)\delta t)$$

$$\Delta\mathbf{v}_{ij} = \sum \Delta\mathbf{R}_{ik}(\mathbf{a}_k - \mathbf{b}_a)\delta t$$

$$\Delta\mathbf{p}_{ij} = \sum \Delta\mathbf{v}_{ik}\delta t + \tfrac{1}{2}\Delta\mathbf{R}_{ik}(\mathbf{a}_k - \mathbf{b}_a)\delta t^2$$

Sliding window ($N = 10$) minimizes visual + IMU residuals via gradient descent.
Limitation: simple optimizer $\rightarrow$ poor scale recovery.

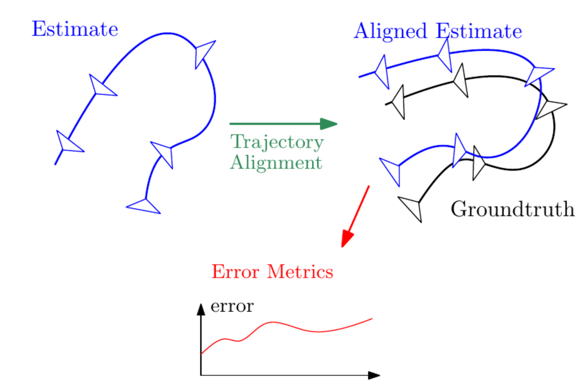


Fig. 2: ORB feature matching

Experimental Setup

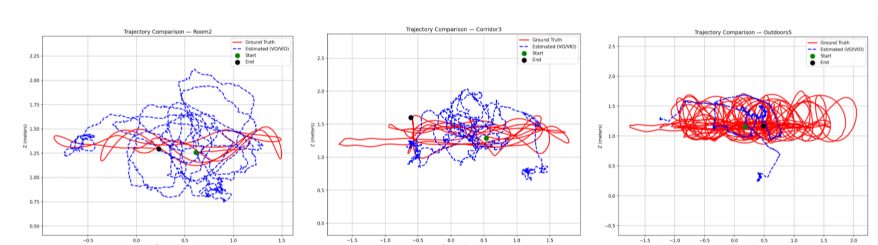
Dataset: TUM VI (Schubert et al.). Sequences: Room2, Corridor3, Outdoors5.
Calibration: $f_x = 190.978, f_y = 190.973, c_x = 254.932, c_y = 256.897$. **Evaluation:** VO $\rightarrow$ Sim(3), VIO $\rightarrow$ SE(3). **Hardware:** Intel CPU, Python 3.12, OpenCV 4.8.

Results — Monocular VO

Sequence	ATE (m)	RPE (m/100m)
Room2	0.7224	0.0150
Corridor3	1.1316	0.0107
Outdoors5	1.0876	0.0107

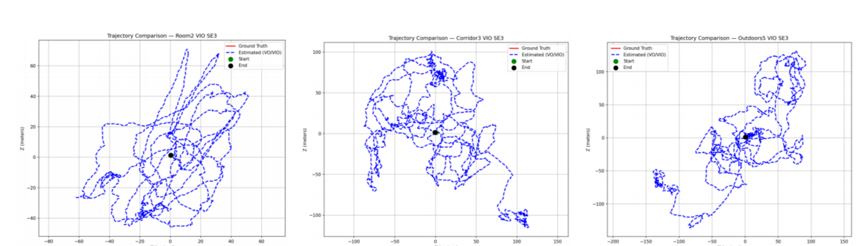
Observation: Low RPE, high ATE $\rightarrow$ local accuracy, global drift. **Start-end drift:** Corridor3 321.81m, Outdoors5 978.22m.

Trajectory Visualizations



Ground truth vs estimated

Fig. 3: VO Trajectory — Room2



Large drift due to missing scale

Fig. 4: VIO Trajectory — Corridor3

Results — VIO & Scale Failure

VIO with SE(3) (no scale correction):

Sequence	ATE (m)	RPE (m/100m)
Room2	0.7564	0.9995
Corridor3	1.0545	0.9996
Outdoors5	0.8673	1.0004

Key finding: RPE ~ 1.0 m/100m $\rightarrow$ metric scale **not recovered**. **Partial success:** Outdoors5 VIO ATE improves from 1.0876m to 0.8673m.

Failure Cases, Ablation & SOTA

Ablation (Room2):

Method	ATE (m)	Time (ms)
Ratio only	0.7564	81.96
+ cross-check	0.7224	137.00

Cross-check improves ATE by 4.5% at $+67\%$ runtime.
Failure cases: fast motion (<8 matches), low texture (sky), scale drift.
Comparison with SOTA (Room2):

Method	ATE (m)
ORB-SLAM3	~ 0.05
VINS-Mono	~ 0.10
Ours (VO)	0.7224
Ours (VIO)	0.7564

Conclusion & Future Work

Conclusions:

- Monocular VO provides accurate local motion (low RPE) but scale ambiguity causes high ATE.
- IMU preintegration alone does NOT guarantee metric scale recovery.
- A proper nonlinear optimizer (Levenberg-Marquardt) and online bias estimation are essential.
- Cross-check matching improves ATE by 4.5%.

Future work:

- Replace gradient descent with Levenberg-Marquardt (OKVIS-style).
- Add online IMU bias correction.
- Implement keyframe selection and marginalization.
- Loop closure for drift reduction.

References

References

[1] Scaramuzza & Fraundorfer, IEEE RAM, 2011.

[2] Forster et al., IMU Preintegration, RSS 2015.

[3] Leutenegger et al., OKVIS, IJRR 2015.

[4] Qin et al., VINS-Mono, IEEE TRO 2018.

[5] Campos et al., ORB-SLAM3, IEEE TRO 2021.